

How to Prevent Active Directory Attacks by Securing Privileged Accounts

 fidelissecurity.com/threatgeek/active-directory-security/securing-privileged-accounts-and-domain-admins

Srestha Roy

February 10, 2026



Key Takeaways

Let's be honest—when Active Directory is compromised, the incident is never small.

Almost every major enterprise breach involves Active Directory at some point. Attackers may enter through phishing, malware, or a misconfigured endpoint, but their real goal is always the same: gain control over privileged identities and Domain Admin accounts.

Once that happens, containment becomes difficult and recovery becomes painful.

Preventing Active Directory attacks isn't about adding more tools. It's about securing the identities that hold the keys to the kingdom. This blog breaks down how [Active Directory attacks](#) actually happen, why privileged accounts are the main target, and what best practices truly reduce risk in real environments.

Why Active Directory becomes the center of enterprise attacks

Active Directory sits at the core of identity, authentication, and authorization. It determines who can log in, what they can access, and which systems trust one another. When attackers gain influence here, they inherit that trust by default.

Active Directory controls enterprise-wide trust

[Active Directory](#) acts as the authoritative source for identity across the environment. Every authentication request, group membership, and access decision depends on it.

This means compromising Active Directory doesn't just give access to one system. It gives attackers the ability to impersonate users, create new identities, and redefine trust relationships across the domain. That level of control is far more valuable than accessing a single application or server.

1. Identity-based access amplifies attacker reach

Modern environments rely heavily on identity-based access rather than network boundaries. Once authenticated, users and services can access multiple systems without re-authenticating.

Attackers exploit this design. Instead of attacking systems one by one, they target identities that already have broad access. When a single identity is compromised, attackers can move laterally using legitimate permissions rather than exploits.

This is why Active Directory attacks often feel invisible in the early stages.

2. Privilege sprawl expands the attack surface

Over time, Active Directory environments accumulate excess privilege. Users retain access they no longer need. Service accounts gain permissions for convenience. Administrative roles are assigned permanently.

Each unnecessary permission becomes a potential attack path. An account that was harmless years ago may now have enough access to escalate privileges if compromised. This sprawl is one of the most common weaknesses in Active Directory security.

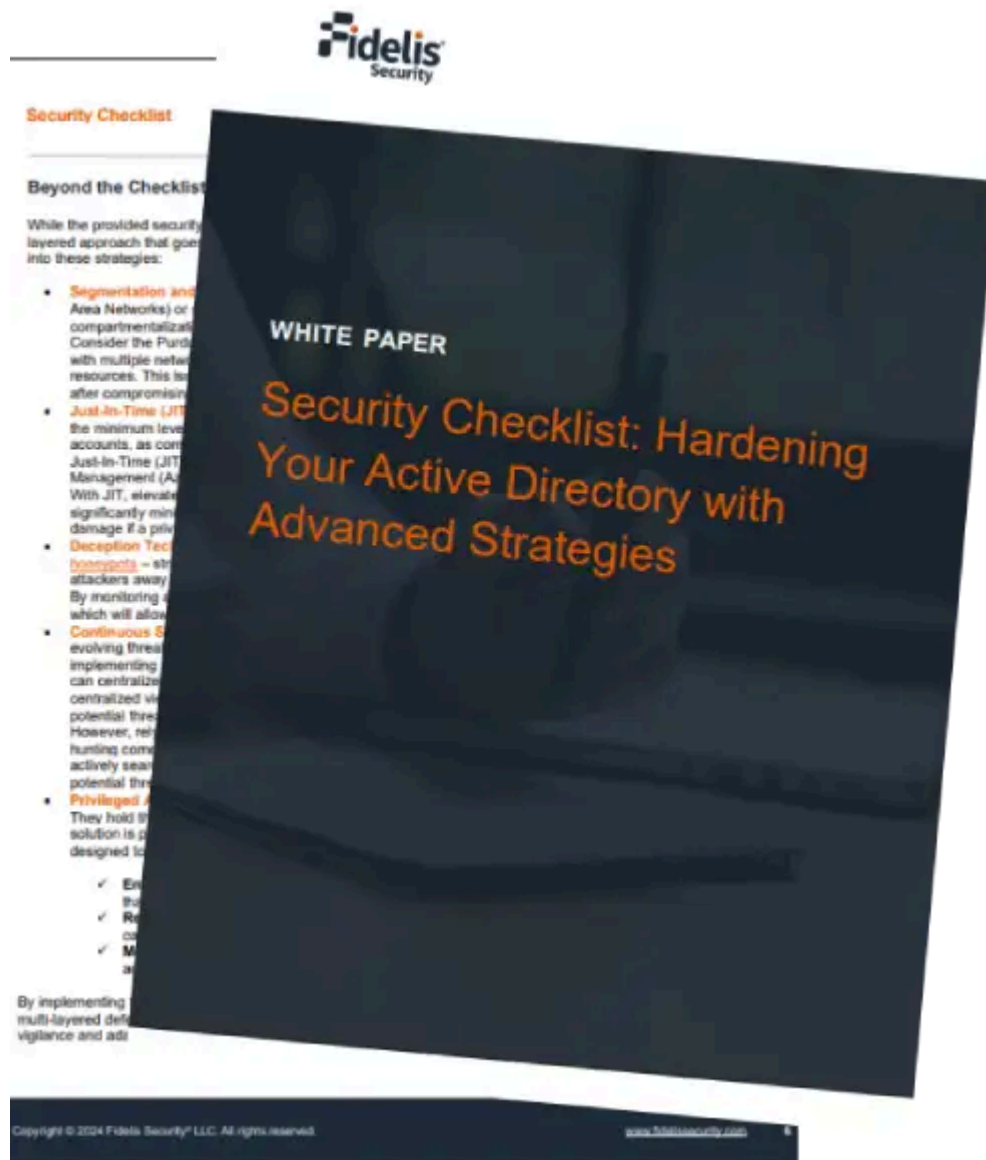
Attackers don't create these paths — they discover and reuse them.

3. Limited visibility hides early warning signs

Active Directory attacks rarely start with Domain Admin access. They begin with subtle changes: unusual logins, privilege usage outside normal patterns, or unexpected access attempts.

Without strong visibility into identity behavior, these early signals are easy to miss. Actions performed using valid credentials often look legitimate, even when they are part of an attack.

This lack of visibility allows attackers to operate quietly until they reach high-value privileges.



How do attackers abuse privileged identities in Active Directory?

1. Standing privileges create persistent attack paths

Many organizations assign permanent admin rights “just in case.” These standing privileges become permanent attack paths.

For example, a user who was granted admin rights for a temporary project may retain those privileges for years. If that account is compromised later, attackers inherit elevated access instantly.

This is one of the most common failures in active directory privileged identity management.

2. Service accounts as overlooked attack vectors

3. Domain Admin accounts as the ultimate objective

4. Abuse of delegated permissions and misconfigurations

What does strong Active Directory security actually require?

1. Least privilege enforced across all roles

2. Active directory privileged identity management in practice

Privileged Identity Management (PIM) replaces standing access with time-bound, approved elevation.

Instead of permanent admin rights, users request access when needed. Access is logged, limited, and revoked automatically.

This significantly reduces the window attackers can exploit.

3. Strong authentication for privileged accounts

4. Visibility into privileged account behavior

How can organizations secure privileged accounts and Domain Admins?

1. Separate admin and user identities

2. Restrict Domain Admin usage

3. Monitor and audit privileged access continuously

4. Protect privileged accounts at the endpoint level

How does this prevent Active Directory attacks in real life?

How Fidelis Security helps you achieve stronger Active Directory protection

- **Spotting Active Directory attacks early, before they spiral**

Most AD attacks don't start with something dramatic. They begin with small, easy-to-miss signs—an odd authentication pattern, a privilege used in an unusual way, a directory action that doesn't quite fit. [Fidelis helps bring these early signals into view](#), so teams can intervene while the attack is still manageable, rather than discovering it after damage has already spread.

- **Making privileged account misuse easier to recognise**

In day-to-day operations, privileged accounts are busy. That makes misuse hard to spot. Fidelis gives teams clearer visibility into how Domain Admins and other high-privilege identities are actually being used, making it easier to tell the difference between routine administrative work and activity that suggests credentials are being abused or privileges are being pushed too far.

- **Breaking attacker movement inside the domain**

Attackers rely on blending in. They move laterally using the same tools and permissions administrators use every day. By [combining Active Directory-aware monitoring with deception](#) techniques, Fidelis helps surface activity that would otherwise pass as normal, allowing teams to interrupt lateral movement and persistence before control of the domain is established.

- **Helping SOC teams respond with confidence, not guesswork**

When an AD incident unfolds, uncertainty slows everything down. Fidelis connects Active Directory signals with network and endpoint context, giving analysts a clearer picture of where the activity began, which identities are involved, and which systems are affected. That clarity makes response faster and, just as importantly, more confident during high-pressure situations.

Datasheet

Fidelis

Multi-layered Defense

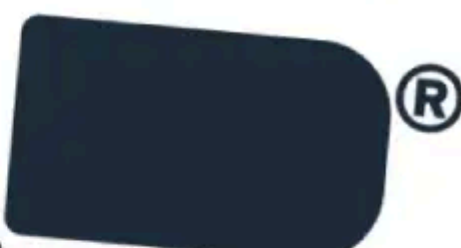
Active Directory (AD) is the enterprise's management. It authenticates and authorizes, provides for the storage and deployment of services such as management, and more. launch point from which it escalates privileges, and execution, data exfiltration.

Protecting AD is a prime target. But many tools fall short of imminent attack. And in game over. They can't network traffic and data protection tools.

That's where Fidelis Active Directory Intercept comes in.

See More. Stop Only with Active Directory Intercept.

Fidelis Active Directory Intercept gives you exactly how, where, into your network, a ability to defend against.



Fidelis Active Directory Intercept™

Multi-Layered Active Directory Defense



One Platform for All Adversaries
